FIG. 1: VOLUMETRIC MASS ESTIMATION, PERISHABILITY KINETICS & SAFETY SHUTOFF

Technical Field: Computer vision container volumetrics, 12-category bulk density tensors, room-temperature shelf-life kinetics, and automated biohazard safety shutoff circuits.

[START] Food Donor Initiates Redistribution Telemetry
Web/Mobile Client Session Initialized; Donor Coordinates Ingested via GPS Geocoding

▼

1. Ingest Batch Telemetry & Dual-Perspective Photography
• Ingests 12 Food Categories, Prepared Timestamp, Geolocation (Lat/Lng), Packaging Type, and Orthogonal Top/Side Photos
◆ **DECISION D1: Is Donor Profile Authenticated? ├── YES: Proceed to Volumetric Vision | └── NO: Issue 24h HMAC-SHA512 JWT Bearer Token**

▼ **[Authenticated Context Established]**

2. Volumetric Computer Vision & Orthogonal Surface Geometry Extraction
• Top-surface perimeter analysis extracts container cross-sectional area (A_surf); side-profile computes fluid fill level (F_eff %)

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◆ D2: [YES] Standard Calibrated Receptacle	◆ D2: [NO] Custom / Irregular Receptacle
• Reference Standard Matrix (5L, 10L, 20L, 50L)	• Monocular Depth & Fill-Level Bounding Volume
• Effective Volume: V_eff = V_box × (Fill% / 100)	• Effective Volume: V_eff = V_custom × (Fill% / 100)
• Repeatability: +5% Boost (89%), Margin: ±8%	• Baseline Model Confidence: 82%, Margin: ±12%

▼ **[Both Branches Converge to Bulk Density Physics Transformation]**

3. 12-Category Food Bulk Density Physics Transformation
• Physical Mass: M = V_eff × ρ (kg) | Density Tensor ρ: Rice (0.80), Curry (0.95), Dairy (1.03), Bakery (0.35), Produce (0.60)
• Portion Count Calculation: Servings S = M / (Grams_per_serving / 1000) (100g - 250g per nutritional portion)

▼

4. Dynamic Perishability Kinetics & Safe Shelf-Life Query
• Query Category Room-Temperature Shelf-Life T_shelf (Dairy 4h, Cooked 5-6h, Bakery 24h, Packaged 168h)
• Elapsed Transit Duration: Δt = now() - t_prepared | Remaining Safe Window: t_rem = max(0, T_shelf - Δt)

▼

◆ D3: [YES] Food Safe (t_rem > 0)	◆ D3: [NO] Bio-Hazard Safety Shutoff
• Organoleptically Sound & Safe for Human Consumption	• Status locked to BIO_DIVERTED; human embargo
• Batch approved for redistribution pipeline	• Divert to anaerobic composting / bio-digestion depot
• Calculate Urgency: P = 0.30·Qty + 0.50·Fresh + 0.20·Time	• Credit Methane Avoidance (IPCC AR6 GWP 28×)

▼ **[Safe Consignments Evaluated for Priority Classification]**

◆ D4: [YES] Emergency Dispatch (P >= 75 | t_rem <= 2h)	◆ D4: [NO] Standard Redistribution Queue
• High-priority WebPush/SMS broadcast to nearby shelters	• Enqueued into regional redistribution cluster pool
• Elevated response SLA: 10-minute acceptance requirement	• Standard 15-minute response SLA window
• Pre-emptively reserve nearest capable volunteer vehicle	• Scheduled batch clustering for multi-stop route solver

[HANDOFF TERMINAL] Validated Consignment Dispatched to Shelter Allocation Engine (FIG. 2)

FIG. 2: PROACTIVE SHELTER DEFICIT FORECASTING & MULTI-CRITERIA NGO ALLOCATION

Technical Field: Time-series statistical demand forecasting, all-pairs Haversine geospatial proximity matrices, multi-objective shelter ranking, and automated SLA cascade escalations.

[START] Ingestion of Validated Food Surplus Consignment from FIG. 1
Consignment Parameters: Mass M (kg), Food Category, Urgency Priority P, Donor Coordinates

▼

6. 24-Hour Regional Forecast Cycle & Recipient Telemetry Ingestion
- Ingests Recipient Shelter Inventory, Bed Capacity, and Dietary Preferences across Target Urban Region
- Ingests Day-of-Week Trend Multipliers (Weekday 1.00× vs. Weekend 1.15×) and Local Festival Multipliers

▼

7. Time-Series Meal Deficit Calculation & Feeding Corridor Demarcation
- Projected Deficit Formula: Deficit = Bed_Capacity × 2.20 × Day_of_Week_Trend × Festival_Multiplier
- Geospatial clustering demarcates high-deficit feeding corridors hours before commercial surplus generation occurs

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8. All-Pairs Haversine Geospatial Distance Matrix Computation
- Compute great-circle arc distance between Donor Node and all candidate Recipient Shelter Nodes:
- D = 2R · arcsin(sqrt(sin^2(Δlat / 2) + cos(lat_donor) · cos(lat_ngo) · sin^2(Δlng / 2)))

▼

9. Multi-Criteria NGO Allocation & Compatibility Ranking
- Objective Scoring Function: S_ngo = (0.40 / D) + (0.35 × Capacity_Fit) + (0.25 × Historical_Acceptance_Speed)
- Rank candidate shelters in descending compatibility order; transmit automated invitation to Rank-1 NGO

▼

◆ D5: [YES] Primary NGO Accepts in SLA (15 min)
- Lock reservation allocation; Status = ACCEPTED
- Assign target destination: Designated_NGO_Id
- Lock inventory allocation to prevent duplicate claiming

◆ D5: [NO] SLA Expired / Declined
- Dynamic Auto-Escalation Cascade Circuit
- Release lock; pop primary NGO from candidate queue
- Re-dispatch consignment to Rank-2 shelter (Loop Step 9)

▼ **[Consignment Accepted: Query & Bind Volunteer Fleet]**

10. Query Active Volunteer Fleet & Bind Transport Assignment
- Filter on-duty volunteers by spatial transit corridor; verify gross payload capacity (C_max >= Consignment Mass M)
- Update State: Pickup Status = ASSIGNED, Transport Driver = Volunteer_Id; transmit automated dispatch notification

[HANDOFF TERMINAL] Consignment Dispatched to CVRP Vehicle Logistics Engine (FIG. 3)

FIG. 3: PERISHABILITY-CONSTRAINED CVRP ROUTING, ZERO-TRUST CUSTODY & SHA-256 AUDIT LEDGER

Technical Field: Operations research Capacitated Vehicle Routing Problem (CVRP), non-deterministic cryptographic optical tokens, digital scale validation, and immutable SHA-256 ledgers.

[START] Multi-Stop Dispatch Request & Volunteer Location Ingestion
Ingestion of Volunteer Location, Active Donor Pickup Nodes, and Destination Shelter Depots

▼

11. Perishability-Constrained Capacitated Vehicle Routing (CVRP) Optimization
- Assemble candidate pickup and delivery nodes; solve route trajectory using Nearest-Neighbor heuristic solver
- Enforce strict chronological topological precedence (every pickup must precede its respective delivery dropoff)

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◆ D6: [YES] Payload Exceeds C_max
- Capacity Overflow Split Resolution
- Schedule mandatory intermediate shelter dropoff
- Bifurcate tour: Leg 1 (Dropoff) -> Leg 2 (Pickup)

◆ D6: [NO] Payload Within Limit
- Consolidated Continuous Logistics Route
- Combine multi-stop pickups into single tour
- 28.5% distance reduction, saves fuel (0.12 L/km)

▼ **[Both Branches Converge into Navigation Manifest & Security Token Generation]**

12. Dispatch Manifest & Generate Non-Deterministic QR Handshake Token
- Transmit optimized turn-by-turn waypoint route manifest to volunteer mobile navigation client
- Generate cryptographically secure single-use token: τ = randomUUID().substring(0,8); render 2D QR code on donor app

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13. Volunteer Origin Physical Custody Transfer & Optical Token Handshake
- Volunteer performs visual inspection & scans donor's optical QR token -> State: PICKED_UP; GPS transit tracking active

▼

14. Destination Shelter Arrival & Digital Scale Mass Check-in
- Measure actual physical mass on shelter digital scale: M_actual (kg); scan optical QR token on container: τ_scan

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◆ D7: [YES] Token Matches (τ_scan == τ_stored)
- Zero-Trust Handshake Authentication Verified
- State Transition: Status = DELIVERED, Pickup COMPLETED
- Closed-loop error: $\varepsilon = (|M_{actual} - M_{est}| / M_{actual}) \times 100$

◆ D7: [NO] Zero-Trust Fraud Lockout Gate
- Reject handover; freeze delivery transition; lock session
- Transmit automated fraud alert & GPS trail to admin
- TERMINAL: Custody Breach Incident Logged

▼ **[Validated Delivery: Recalibrate AI & Commit Cryptographic Ledger]**

15. Closed-Loop AI Recalibration & Immutable SHA-256 Ledger Commitment
- Recalibrate Volumetric Model with Scale Ground-Truth (Historical MAPE = 3.8%)
- IPCC Nexus: CO_2e (1.9 kg/kg), Water (1400 L/kg), Meals (M / 0.5 kg) | Canonical Digest: SHA-256('EVS-LEDGER-V1|...')

[TERMINAL] Food Redistribution Completed, Cryptographically Audited & Public ESG Certificate Issued